Team Control Number

**83066**

Problem Chosen

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**MCM/ICM Summary Sheet**

# How Many Languages?

## Summary

In this problem, we are required to build a mathematical model to investigate the trends of global languages and to help a multinational service company to become more international by looking for location options of new offices.

In order to achieve this goal, we need to investigate the use condition of languages in different countries. To simplify the description, here we take English for example. We divide the problem into two parts, the number of people speaking English in English-speaking countries and the number of people speaking English in non-English speaking countries. The previous number is associated with migration, population and proportion of people whose mother tongue is English. And the latter number is associated with language vitality, government support and education popularity.

We adopt the Analytic Hierarchy Process to calculate languages' vitality. The vitality of language is determined by many factors. With AHP, we can work out the weight of each factor and the exact value of language vitality. To calculate the number of people speaking English in non-English speaking countries, we use the multiple linear regression model to analyze the quantitative relationship among language vitality, government support and education popularity. We adopt GM (1, 1) to make predictions, such as the future values of language vitality, government support and education popularity. With the three models and another two algorithms working together, we manage to solve the original problem. After modeling, we make some model assessment methods to make sure that the results we obtain from modeling are reasonable.

Through analysis and calculation, we draw a set of thermodynamic diagrams which describe the geographic distribution of various language speakers over time. We make a conclusion that the number of native speakers and total language speakers is getting larger and larger over time. The languages in the current top-ten list will not be replaced in a few years. Global population and human migration patterns do affect geographic distributions.

Taken many factors into account, we provide some reasonable advice to help international companies expand their business by opening new offices.

**Key Words: Languages, Analytic Hierarchy Process, Gray Model, Clustering**

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# 1 Introduction

## 1.1 Background

Currently, there are about 6900 languages around the world. But about half of the world's population speaks one of the following ten languages as their mother tongue: Mandarin (incl. Standard Mandarin), Spanish, English, Hindi, Arabic, Bengali, Portuguese, Russian, Punjabi, and Japanese.

However, many of these people also speak a second language. When taking the total number of speakers of a particular language into account, the languages' rank changes from that of native language provided.

The total number of speakers of a language may increase or decrease over time because of a variety of influences. Those influences include, but not limited to, the languages used or promoted by their government, the languages used in schools, social pressures, migration and assimilation of cultural groups, and immigration and emigration with countries that speak other languages.

Moreover, in our globalized, interconnected world there are additional factors that allow languages that are geographically distant to interact. These factors include international business relations, increased global tourism, the use of electronic communication and social media, and the use of technology to assist in quick and easy language translation.

## 1.2 Restatement of the Problem

In this problem, we are required to build a mathematical model to investigate the trends of global languages and to help a multinational service company to become more international by looking for location options of new offices. The whole problem can be decomposed into three parts:

- Model the distribution of various language speakers over time and make some predictions about variation trends accordingly.
- Based on the modeling, help the company to locate some international offices to not only maximize the benefits but also save their resources.
- Write a memo to summarize the results and recommendations.

## 1.3 Notions and Assumptions to Notice

- Language Vitality

  We know that languages expand and shrink on the back of the social, cultural, military, scientific, technological, organizational and other strengths and weaknesses of countries using them as official languages. We define this attribute of language as **language vitality**.
- Official Language and Mother Tongue

  For countries we will refer to in future paragraphs, the UK, Russia, Germany, Indonesia, Algeria, Spain, Japan and Brazil, the official language of these countries and the mother tongue of majority people in these countries are the same. We assume the official language and the mother tongue to be the same language in this article.
- Factors Taken into Account

  In order to make our modeling and prediction more accurate, **we will take more factors into account**. These factors also affect the distribution of people using a certain language, such as population growth rate, level of national education, technological influence, cultural influence and the economic power of countries.

Meanwhile, **we assume the factors taken into account should be mutually exclusive**. As a result, to make the analysis process more succinct, **we will omit factors which are redundant or less important**. Let's take some examples. As social pressure is highly associated with the economic power of the country, we will omit the discussion of social pressure.

- Using a single indicator to denote a factor

  Factor like economic power is associated with many factors, such as GDP, GNP and so on. To simplify the problem, we use a single indicator to denote each factor mentioned above. The following table shows which indicator selected to denote each factor.

| Factor | Economic Power | Cultural Influence | Technological Influence | Development of Tourism | Migration Rate |
|---|---|---|---|---|---|
| Indicator | Gross Domestic Product (GDP) | Number of national feature films produced (per year) | Number of researchers in Research & Development (per million people) | Number of arrivals of international tourism (per year) | Number of people moving in (per year) |

Table 1: The Indicators of Language Vitality

## 1.4 Overview of Study Process

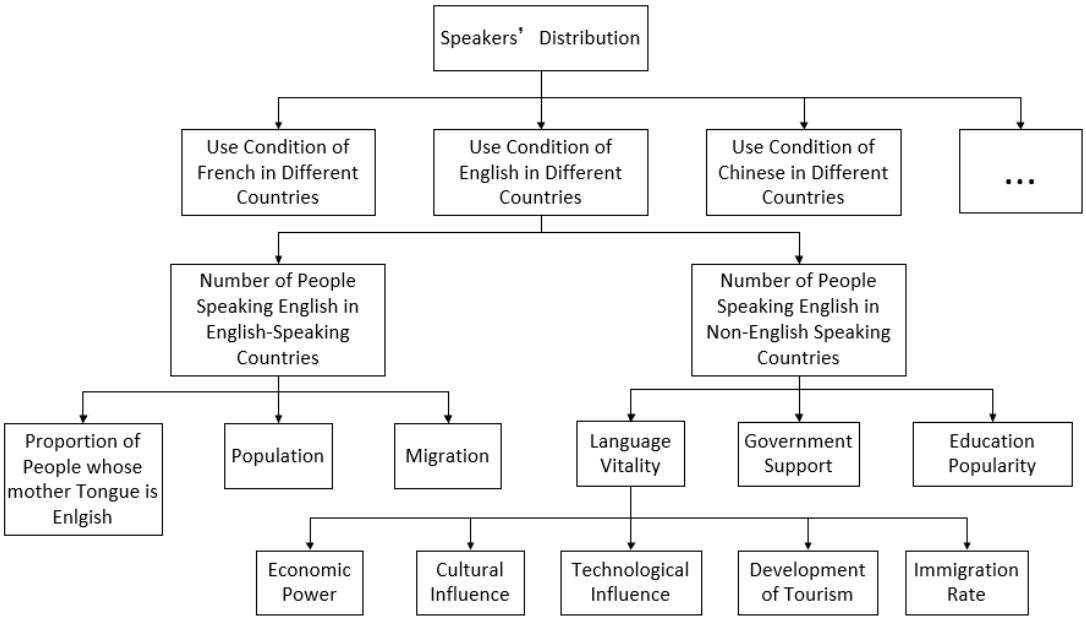


Figure 1: Structure of Speakers' Distribution

Here is the interpretation of the figure above. Let's read the figure top-down.

In order to get the geographical distribution of various language users, we decide to sample and investigate some of the countries around the world. By analyzing each of the countries' use of various languages, the distribution of speakers in the global range can be summarized.

In the theoretical system we establish, we divide the problem shown on level 2 of the figure into two different conditions. They are the use of the mother tongue and the use of other languages in different countries.

We assume that during a certain period of time, the number of native speakers in one country is related to the following factors: population growth rate, number of people moving in and moving out.

Meanwhile, the number of people using non-native language is related to three factors: the vitality of the language itself, support of government and the level of national education in this country.

Among the three factors, language vitality is a complex one. It is determined by five more subfactors. The subfactors are, the economic power, cultural influence, technological influence, development of tourism and immigration rate of countries using the language as their mother tongue.

### 1.5 Literature Review

Chunhui Wang proposed a concept as referring to 'Language Order' which means the structure and pattern of language. In his paper *Language Order in Contemporary World*, he concluded that the contemporary human language presents a state of bi-directional polarization between major languages and endangered languages.

Fishman analyzed the expansion of English, the continuous function of local languages, the sustainability of minority languages. Then he made prospects for future language development.

Abram de Swaan compared five different languages through a core concept called Q-value from aspects including political science, economics and sociology. He described the process in which a language become a popular one.

## 2 Data Preprocessing

### 2.1 Data Reconstruction

In many cases, the data we find from websites or other resources are not complete. When we look for data about the foreign-born population of the USA from 2005 to 2015, actually we can only find the data of 2005, 2010 and 2015.

| Year | 2005 | 2010 | 2015 |
|---|---|---|---|
| Foreign population (Number of People) | 39258293 | 44183643 | 46627102 |

Table 2: Foreign Population of the USA in 2005, 2010, 2015

We are required to reconstruct the unknown values by analyzing the known values. Here we adopt the linear interpolation method. By calculating the average value of two known values, we can obtain the value of an unknown one between the two known values. In the immigration example, we can get data for each year from 2005 to 2015 by using the method several times.

### 2.2 Data Standardization

In some practical problems, the sample data we get are multi-dimensional, that is, a sample is characterized by multiple features. It is clear that the dimension of features and the magnitude of numerical values are not the same. Consequently, if we directly put the raw data into the calculation, the impacts of various features on the result can be unreasonable. By standardization, different features can be made to have the same scale and their impacts on the result are reasonable. After the raw data being divided by the standard deviation, we obtain the standardized data.

# 3 Models

## 3.1 Overview

- Analytic Hierarchy Process (AHP)
  Firstly, we adopt AHP to calculate languages' vitality. The vitality of language is determined by many factors. In this model, we regard the language vitality as the target layer, and regard the various factors as the criteria layer.
- Multiple Linear Regression Model
  Multiple linear regression model aims to analyze the quantitative relationship between objectives. In our article, we use this model to reveal the relationship between language vitality, government support and education popularity.
- Gray Model, GM (1, 1)
  The gray system seeks for the changing rule by collation of original data, which is a way to find data's real rule based on original data. Since GM (1, 1) can obtain accurate predictions based on a small quantity of data which exactly satisfy our requirements, we decide to adopt GM (1, 1) to make predictions, such as the future values of language vitality, government support and education popularity.
- Gray Relational Prediction Analysis
  Because of the complex relationship among the tree indicators, we choose Gray Relational Prediction Analysis to confirm the selection system.
- Maximum Minimum Distance Algorithm
  The Maximum Minimum Distance Algorithm is a heuristic clustering algorithm, which is based on the calculation of Euclidean distance.

## 3.2 Symbols

- $Econ_{Cont}^{Yr}$
  Economic power indicator of a country 'Cont' in year 'Yr'.
- $Cutl_{Cont}^{Yr}$
  Cultural influence indicator of a country 'Cont' in year 'Yr'.
- $Tech_{Cont}^{Yr}$
  Technological influence indicator of a country 'Cont' in year 'Yr'.
- $Tour_{Cont}^{Yr}$
  Tourism indicator of a country 'Cont' in year 'Yr'.
- $Immg_{Cont}^{Yr}$
  Immigration rate indicator of a country 'Cont' in year 'Yr'.
- $Contb_{Cont}^{Yr}$
  The contribution of a country 'Cont' to language vitality in year 'Yr'.
- $LanV_{Lan}^{Yr}$
  Language vitality indicator of a language 'Lan' in year 'Yr'.
- $CER_{Cont}$
  College enrollment ratio in country 'Cont'.
- $GovS_{Cont}$
  Government support of a certain kind of language in country 'Cont'.

- $Propt_{Cont,Lan}$
  Proportion of people speaking language 'Lan' in country 'Cont'.
- $Propt_Natv_{Cont,Lan}^{Yr}$
  Proportion of people speaking language 'Lan' as their mother tongue in country 'Cont' in year 'Yr'.
- $Propt_NNatv_{Cont,Lan}^{Yr}$
  Proportion of people speaking language 'Lan' as their non-mother tongue in country 'Cont' in year 'Yr'.
- $Popl_{Cont}^{Yr}$
  The population of country 'Cont' in year 'Yr'.
- $NumP_Natv_{Cont,Lan}^{Yr}$
  Number of people in country 'Cont' using language Lan as their mother tongue in year 'Yr'.
- $NumP_NNatv_{Cont\ i,Lan}^{Yr}$
  Number of people in country 'Cont' using language Lan as their non-mother tongue in year 'Yr'.
- $NumP_Natv_{Lan}^{Yr}$
  Number of people in country 'Cont' using language Lan as their mother tongue in year 'Yr'.
- $NumP_NNatv_{Lan}^{Yr}$
  Number of people in country 'Cont' using language Lan as their non-mother tongue in year 'Yr'.
- $NumP_{Cont,Lan}^{Yr}$
  Number of people in country 'Cont' using language Lan as their non-native language in year 'Yr'.
- $NumP_{Lan}^{Yr}$
  Number of people speaking language 'Lan' around the world in year 'Yr' (We select a few countries as sample).

## 3.3 Analytic Hierarchy Process (AHP)

The number of people using a certain kind of language is determined by its language vitality. George Weber analyzed some popular languages' vitality on account of 6 factors. It turned out that English ranked first with a score of 37.

Analogously to George Weber, we score the vitality of languages with a series of criterion. The criteria include the number of countries using the language as their mother tongue, those countries' economic power, their cultural influence, their technological influence, their development of tourism and immigration rate.

Consequently, for a language used by a single country as its mother tongue, we only need to measure the comprehensive strength of this country to get the global vitality of the language. As for a language used by multiple countries as mother tongue, its vitality is determined by those countries collectively. We can gain the language's global vitality by accumulating the scores contributed by the countries who speak it.

Let's take English for example. The advantages of developed countries in economic, cultural, scientific and technological aspects, are the basic prerequisites for establishing and maintaining the international status of English.

While evaluating the vitality of a language, all the criteria need to be considered and we should avoid making subjective judgments. As a result, we choose to use AHP to make the decision.

As we have stated above, we have 5 criteria which contribute to the language vitality.

- ✓ Criterion 1: The countries' economic power
- ✓ Criterion 2: The countries' cultural influence
- ✓ Criterion 3: The countries' technological influence
- ✓ Criterion 4: The countries' development of tourism
- ✓ Criterion 5: The countries' immigration rate

**1. Establish a Hierarchical Model**

We need AHP to help us to calculate the weight of each criterion. With the weight of each criterion worked out, the abstract problem is transferred to a mathematics one.

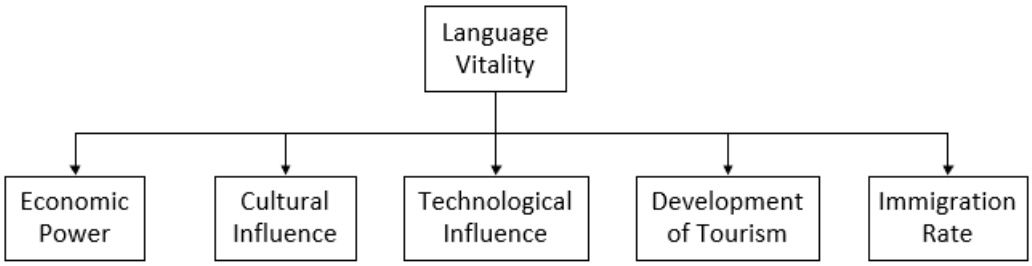


Figure 2: The Factors of Language Vitality

**2. Establish Pairwise Comparison Criteria Judging Matrix**

To begin with, we should determine the pairwise comparison criteria judging matrix. In AHP, 5 methods are used to represent the relative importance between criteria.

The Power Language Index (PLI) is a systematic way of evaluating the vitality and reach of languages using 20 indicators classified into 5 categories which correspondingly match the 5 criteria we mentioned above. According to PLI, then we get the weight of each criterion as is shown in the following table.

| | Economic Power | Cultural Influence | Technological Influence | Development of Tourism | Immigration Rate |
|---|---|---|---|---|---|
| Economic Power | 1 | 3 | 3 | 5 | 7 |
| Cultural Influence | 1/3 | 1 | 1 | 3 | 5 |
| Technological Influence | 1/3 | 1 | 1 | 3 | 5 |
| Development of Tourism | 1/5 | 1/3 | 1/3 | 1 | 3 |
| Immigration Rate | 1/7 | 1/5 | 1/5 | 1/3 | 1 |

Table 3: Pairwise Comparison Criteria Judging Matrix

Note that the judging matrix $\mathrm{A} = (a_{ij})_{n*n}$ where the elements must satisfy

$$a_{ij} > 0,\ a_{ii} = 1,\ a_{ij} = \frac{1}{a_{ji}}$$

**3. Obtain the Criteria Weight Vector**

Then we calculate the eigenvalues and the corresponding eigenvectors. We select the eigenvector $\mathrm{u} = (u_1,\ u_2, \dots, u_n)^T$ with the maximum eigenvalue $\lambda_{max}$ as the criteria weight vector and normalizing it as

$$u_i' = \frac{u_i}{\sum_{j=1}^{n} u_j}$$

As the calculation turns out, the criteria weight vector is

$$u = (0.4691, 0.2001, 0.2001, 0.0862, 0.0427)^T$$

**4. Apply Consistency Checking**

We calculate the consistency ratio by CR=CI/RI. If the calculating ratio is less than 0.1, then we think the judging matrix is qualified and the eigenvector with the largest eigenvalue can be used as the weight vector. If not, we need to modify the judging matrix until the less-than condition is satisfied.

$$\mathrm{CI} = \frac{\lambda_{max} - n}{n - 1}$$

| N | 1 | 3 | 5 | 7 | 9 |
|---|---|---|---|---|---|
| RI | 0 | 0.58 | 1.12 | 1.32 | 1.45 |

Table 4: Random Consistency Indicators

Using the pairwise-comparison judging matrix above in respect of the five criteria. We can calculate coincidence indicator as CI = 0.0317 and the consistency ratio as CR = 0.0283 which indicate our judging matrix is qualified and the eigenvector can be used as the weight vector.

**5. Calculate Language Vitality**

The following formula is used for calculating $LanV_{Lan}^{Yr}$.

$$Contb_{Cont}^{Yr} = u_{Econ} \times Econ_{Cont}^{Yr} + u_{Cult} \times Cult_{Cont}^{Yr} + u_{Tech} \times Tech_{Cont}^{Yr} + u_{Tour} \times Tour_{Cont}^{Yr} + u_{Immi} \times Immi_{Cont}^{Yr}$$

$$LanV_{Lan}^{Yr} = \sum_{i}^{n} Cont_{Cont\ i}^{Yr}$$

Here we take English for example again and calculate the vitality of English. Let $Contb_{Brt}^{2013}$ denote the contribution of British to English in the year 2013, and $LanV_{Eng}^{2013}$ denote the language vitality indicator of English in the year 2013.

To simplify the calculation, here we consider the contribution degree of only five countries to English because they are the most representative countries whose mother tongue is English. They are the USA, Canada, Australia, the UK, and New Zealand.

Below is the raw data. Note that before putting it into the calculation, we should standardize it to the indicator.

| | **Economic Power** / GDP (100 million USD) | **Cultural Influence** / Number of films produced | **Technological Influence** / Number of researchers (per million people) | **Development of Tourism** / Number of tourists (10000 persons) | **Migration Rate** / Number of people |
|---|---|---|---|---|---|
| **USA** | 167681 | 738 | 4271.1 | 6977 | 45907420 |
| **CAN** | 18390 | 93 | 4521.2 | 1659 | 751428 |
| **AUS** | 15604 | 26 | 4310.6 | 638 | 6425909 |
| **UK** | 26782 | 241 | 4055.1 | 3117 | 8346451 |
| **NZL** | 1884 | 25 | 3753.2 | 263 | 1004587 |

Table 5: Contribution Degrees of Five Countries

Now the formula can be written as

$$Contb_{USA}^{2013} = u_{Econ} \times Econ_{USA}^{2013} + u_{Cult} \times Cult_{USA}^{2013} + u_{Tech} \times Tech_{USA}^{2013} + u_{Tour} \times Tour_{USA}^{2013} + u_{Immi} \times Immi_{USA}^{2013}$$

$$LanV_{Eng}^{2013} = Cont_{USA}^{2013} + Cont_{CAN}^{2013} + Cont_{AUS}^{2013} + Cont_{UK}^{2013} + Cont_{NZL}^{2013}$$

The contribution degrees of these countries to English in 2013 are 0.6203, 0.1237, 0.1484, 0.1847 and 0.0749 respectively. Then we can obtain the language vitality of English calculated as 1.1520.

We draw a thermodynamic diagram shows how greatly the five countries contribute to English. The brighter the color is, the greater the contribution is.

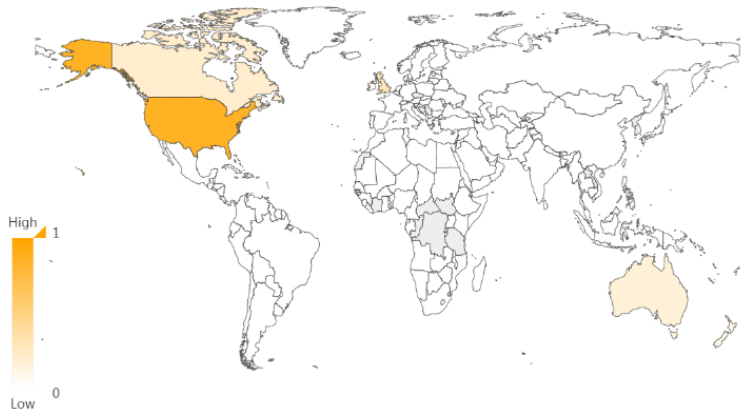


Figure 3: Thermodynamic Diagram about Contribution to English

## 6. Language Vitality of Six Top Languages

By repeating the process above, we can obtain the language vitality of six top languages from 2005 to 2013 and draw a line chart accordingly.

From the line chart, we can conclude that English has always been the most vigorous language around the world. Though the changing trend of English vitality keeps getting down since 2010, there is still no doubt that English will keep ranking number one in the next few years.

Languages ranked top six except for English share the similar vitality. But there is one thing worthy notice, that is, the changing trend of Mandarin vitality keeps going up.

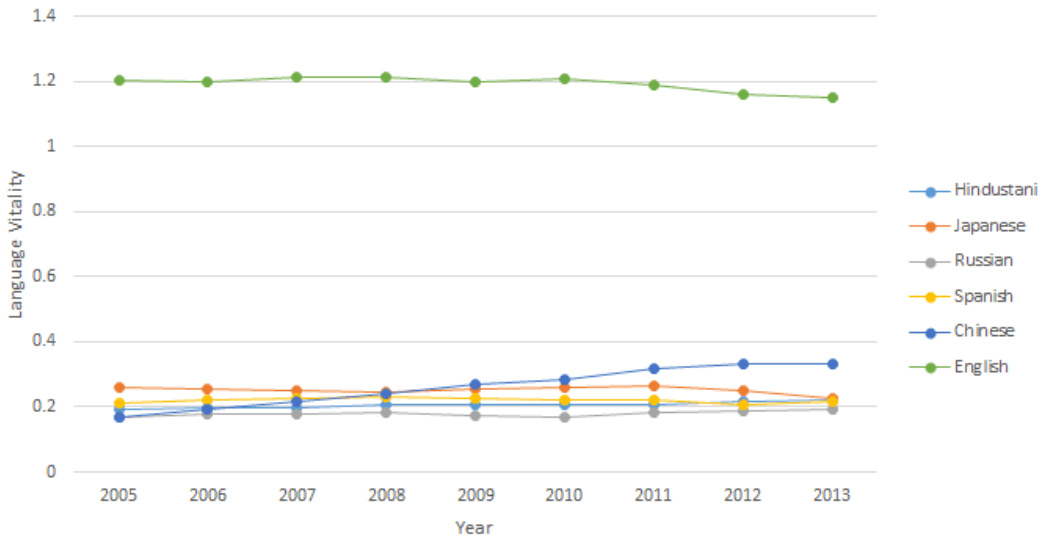


Figure 4: Language Vitality

## 3.4 Multiple Linear Regression Model

**1. Calculate the Coefficients Used for Prediction about Number of Total Language Speakers**

After using AHP to calculate the language vitality, we can further predict the proportion and number of people speaking the language according to language vitality, government support and education popularity.

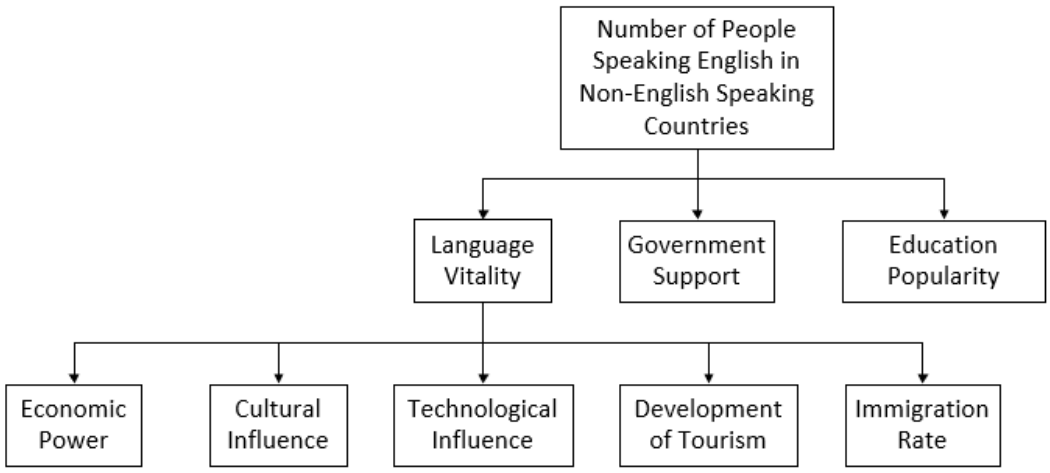


Figure 5: Process Structure

The three factors mentioned above are of different significance when used to calculate the proportion. Similar to the calculation of language vitality, firstly we need to know the weight of each factor. To get the weights, we are required to analyze data about language vitality, government support, education popularization and proportion of people speaking the language in a country over the last few years. When we get the weight of each factor, then we can predict the proportion over time in the future.

Government support is represented by decimal points ranging from 0 to 1. The greater the number is, the higher the government's support is. The degree of education popularity is expressed by college enrollment ratio. In this practical problem, there are three variables in the same process, they both serve and restrict each other. The relationship between them is not completely stable.

Multiple linear regression model aims to analyze the quantitative relationship between objectives, which is widely used in the study of social and economic phenomena. Consequently, we use this model to reveal the relationship between the three variables. The relationship among the four variables can be described as a ternary linear regression equation

$$\widehat{Propt_{Cont,Lan}} = b + a_1 \times LanV_{Lan} + a_2 \times GovS_{Cont} + a_3 \times CER_{Cont}$$

As we describe the detailed process of calculation, we will take China for example and discuss the English vitality in China. We will directly show the conclusions for other cases.

1) Collect and Standardize Raw Data

The vitality of English is obtained from the previous process. We find the other three lines of data on websites of the World Bank and the United Nations Statistics Division. Before putting these data into the calculation, we are required to do data standardization to make the result reasonable. The data we've found are listed as follows.

| Year / Factor | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 |
|---|---|---|---|---|---|---|---|---|---|
| $LanV_{Lan}$ | 1.204 | 1.198 | 1.211 | 1.212 | 1.197 | 1.209 | 1.189 | 1.160 | 1.152 |
| $GovS_{Cont}$ | 0.713 | 0.725 | 0.741 | 0.858 | 0.834 | 0.847 | 0.879 | 0.882 | 0.866 |
| $CER_{Cont}$ | 20.3 | 21.6 | 22.9 | 22.7 | 24.5 | 25.9 | 26.3 | 26.7 | 29.7 |
| $Prop_{Cont}^{Lan}$ | 0.233 | 0.247 | 0.262 | 0.281 | 0.282 | 0.290 | 0.298 | 0.305 | 0.313 |

Table 6: Factor Data

2) Calculate Coefficients

The relationship among language vitality, government support, education popularity and proportion of people speaking English in China can be described as

$$\widehat{Propt_{CHN,Eng}} = b + a_1 \times LanV_{Eng} + a_2 \times GovS_{CHN} + a_3 \times CER_{CHN}$$

In this equation, $\widehat{Propt_{CHN,Eng}}$, $LanV_{Eng}$, $GovS_{CHN}$ and $CER_{CHN}$ are variables representing known data. By putting different sets of data into the equation, we can work out the value of b, $a_1$, $a_2$ and $a_3$. By putting the values above into the equation, we can work out the values of the coefficients b, $a_1$, $a_2$ and $a_3$.

| Coefficient | b | $a_1$ | $a_2$ | $a_3$ |
|---|---|---|---|---|
| Weight | 0.279000 | 0.000993 | 0.015053 | 0.013382 |

Table 7: Weight of Parameters

## 3.5 Gray Model, GM (1, 1)

In this part, we are required to model the distribution of various language speakers over time. To achieve this goal, firstly we need to predict the use condition of various languages in different countries. We take English for example and divide the problem into two parts, the number of native speakers and the number of non-native speakers in different countries.

The gray system seeks for the changing rule by collation of original data, which is a way to find data's real rule based on the original data. Since GM (1, 1) can obtain accurate predictions based

on a small quantity of data which exactly satisfies our requirements, we decide to adopt it.

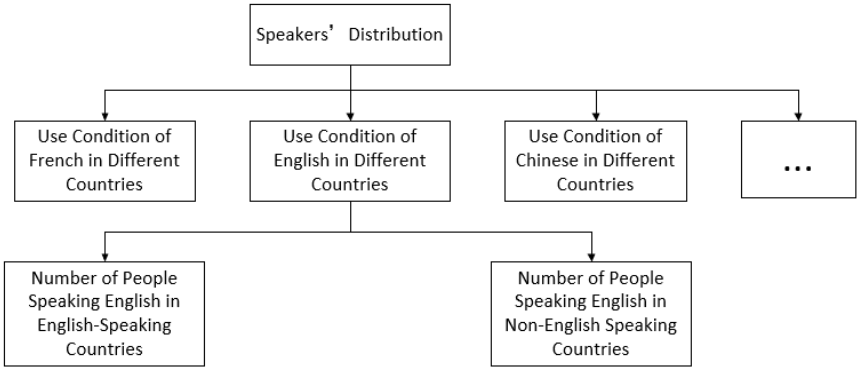


Figure 6: Process Structure

**1. Prediction on Number of People Speaking English in English-Speaking Countries**

In this step, we directly predict the number of people using English as their native language in the next fifty years through the data from 2005 to 2013. We've made the following assumptions: the number of people speaking English in English-Speaking country is determined collectively by population, migration and proportion of people whose mother tongue is English.

Migration population is a small portion of the total population. We assume the impact of migration is so small that we can ignore it. Consequently, the number we evaluate is determined only by the population and proportion of people whose mother tongue is English.

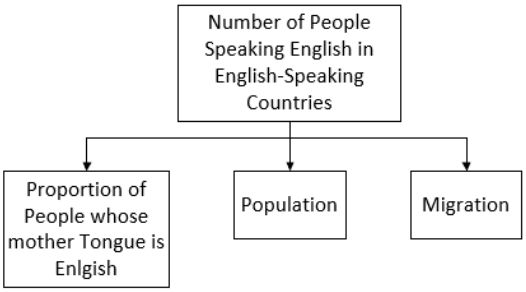


Figure 7: Process Structure

Based on the past proportions, we can predict its changing trends in the next fifty years. The figure below describes the proportion of people whose mother tongue is English in the UK.

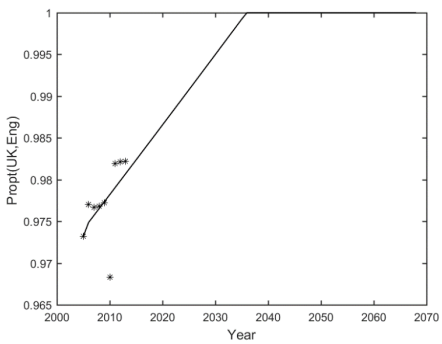


Figure 8: Proportion of People Speaking English in the UK

Then we find the population of the UK in the next fifty years from the website.

| Year | 2020 | 2030 | 2040 | 2050 | 2060 | 2070 |
|---|---|---|---|---|---|---|
| Population | 67334208 | 70578614 | 73125233 | 75381137 | 76971438 | 78211698 |

Table 8: Population

With the population and proportion of people whose mother tongue is English, we can calculate the exact number of people using English as their mother tongue in the next fifty years in the UK.

$$NumP_Natv_{UK,Eng}^{Yr} = Propt_Natv_{UK,Lan}^{Yr} \times Popl_{UK}^{Yr}$$

With the numbers of the UK, the USA, Canada, New Zealand and Australia, we can obtain the total number of people who speak English as their mother tongue in English speaking countries.

$$NumP_Natv_{Eng}^{Yr} = \sum_{i=0}^{n} NumP_Natv_{Cont\ i,Eng}^{Yr}$$

The calculation of other languages follows the same routine.

**2. Prediction on Number of People Speaking English in Non-English-Speaking Countries**

In this step, we need to predict the number of people using English as their non-native language in the next fifty years through the data from 2005 to 2013. We've made the following assumptions: the number of speakers using English as their Non-native language is determined collectively by English vitality, government support and education popularity.

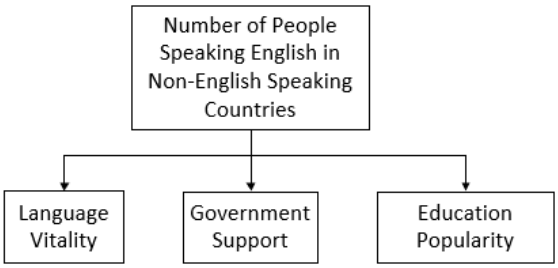


Figure 9: Process Structure

The exact values of the three factors in the next fifty years are still unknown. So we need to adopt a method to predict future values based on previous values. In this case, we've found data of nine years from 2005 to 2013. But we are required to predict future values for the next fifty years.

Once we get the future data, we can use the following formula to calculate the proportion of people using a language in a country. We assume that the weights of the three factors don't change over time.

$$Propt_{Cont,Lan} = 0.279000 + 0.000993 \times LanV_{Lan} + 0.015053 \times GovS_{Cont} + 0.013382 \times CER_{Cont}$$

Then we are able to calculate the number of people using language 'Lan' in country 'Cont' by adopting formula:

$$NumP_{Cont,Lan}^{Yr} = Propt_{Cont,Lan} \times Popl_{Cont}^{Yr}$$

Now let us take English spoken in China as an example to show the detailed steps solving the problem as follows.

**1) Predict Future Values of Language Vitality, Government Support and Education Popularity**

We predict the future values of the three factors by data from 2005 to 2013.

| Year / Factor | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 |
|---|---|---|---|---|---|---|---|---|---|
| $LanV_{Eng}$ | 1.2035 | 1.1984 | 1.2111 | 1.2122 | 1.1968 | 1.2090 | 1.1885 | 1.1600 | 1.1520 |
| $GovS_{CHN}$ | 0.713 | 0.725 | 0.741 | 0.858 | 0.834 | 0.847 | 0.879 | 0.882 | 0.886 |
| $CER_{CHN}$ | 0.203 | 0.216 | 0.229 | 0.227 | 0.245 | 0.259 | 0.263 | 0.267 | 0.297 |

Table 9: Factor Data

The following three figures show the changing trends of English vitality, government support and education popularity in China.

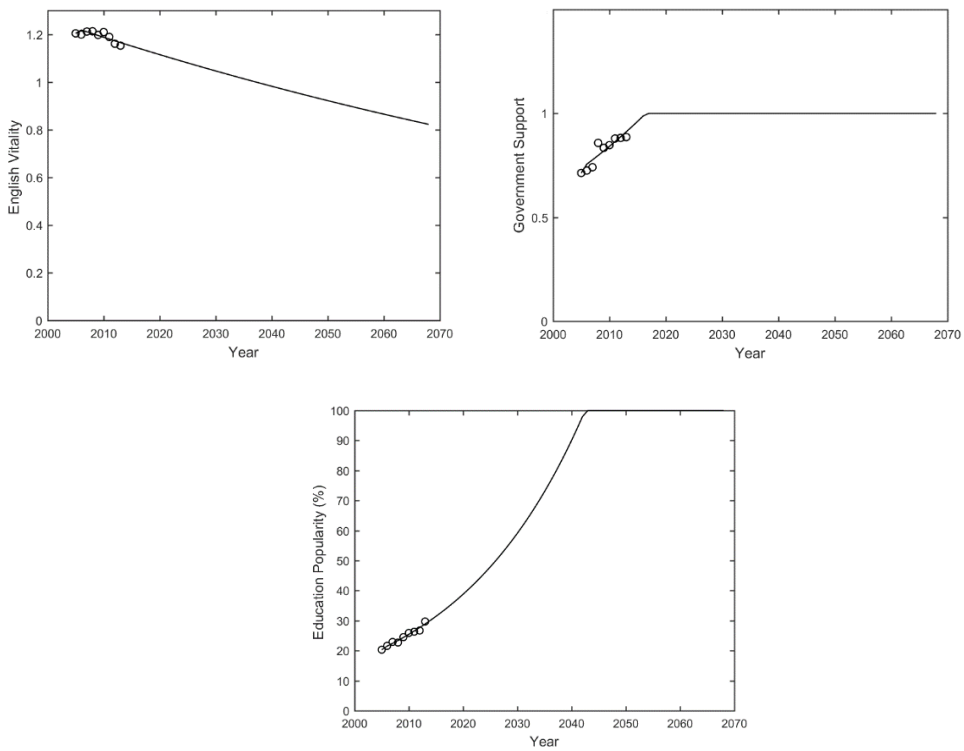


Figure 10: Prediction of Changing Trends

**2) Calculate the Number of People Speaking English in China**

Since we've predicted the changing trends of English vitality, government support and education popularity in China in the future, we can predict the proportion of people speaking English in China according to the formula.

$$Propt_{Cont,Lan} = 0.279000 + 0.000993 \times LanV_{Lan} + 0.015053 \times GovS_{Cont} + 0.013382 \times CER_{Cont}$$

The following figure describes how the proportion of people speaking English in China will change.

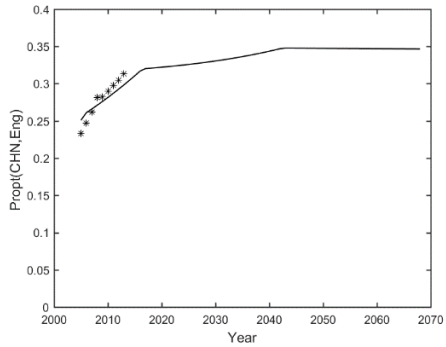


Figure 11: Prediction of Changing Trends

With the proportion, we can calculate the total number of people speaking English in China by

$$NumP_{CHN,Eng}^{Yr} = Propt_{CHN,Eng}^{Yr} \times Popl_{CHN}^{Yr}$$

With the population and proportion of people who speak English as non-mother langue, we can calculate the exact number of people using English as their non-mother tongue in the next fifty years in China.

$$NumP_NNatv_{CHN,Eng}^{Yr} = Propt_NNatv_{CHN,Eng}^{Yr} \times Popl_{CHN}^{Yr}$$

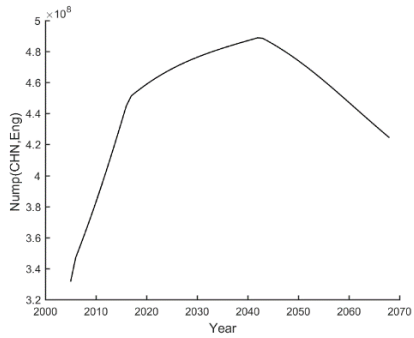


Figure 12: Prediction of Changing Trends

$$NumP_Natv_{Eng}^{Yr} = \sum_{i=0}^{n} NumP_Natv_{Cont\ i,Eng}^{Yr}$$

To simplify the calculation, we just take some of the countries into account, we can obtain the total number of people who speak English as their mother tongue in English speaking countries.

The calculation of other languages follows the same routine.

## 3. Thermodynamic Map of English Global Distribution

We draw a thermodynamic map to describe the distribution of English speakers over time. The three maps are distributions of 2013, 2040 and 2068 respectively. The size of the circle on the map represents the size of the total number of English speakers in that region. To simplify the process, we choose thirteen of all the countries to analyze.

We can learn from that the USA and China have the largest group of English speakers (There

are many English learners in China). **We can also see that the number of native speakers is getting larger and larger over time, especially the number of the USA. Besides, the number of total language speakers is also getting larger.**

**Global population and human migration patterns do affect geographic distributions.** As we analyzed above, there are many other factors which will affect the geographic distribution of languages together with global population and human migration.

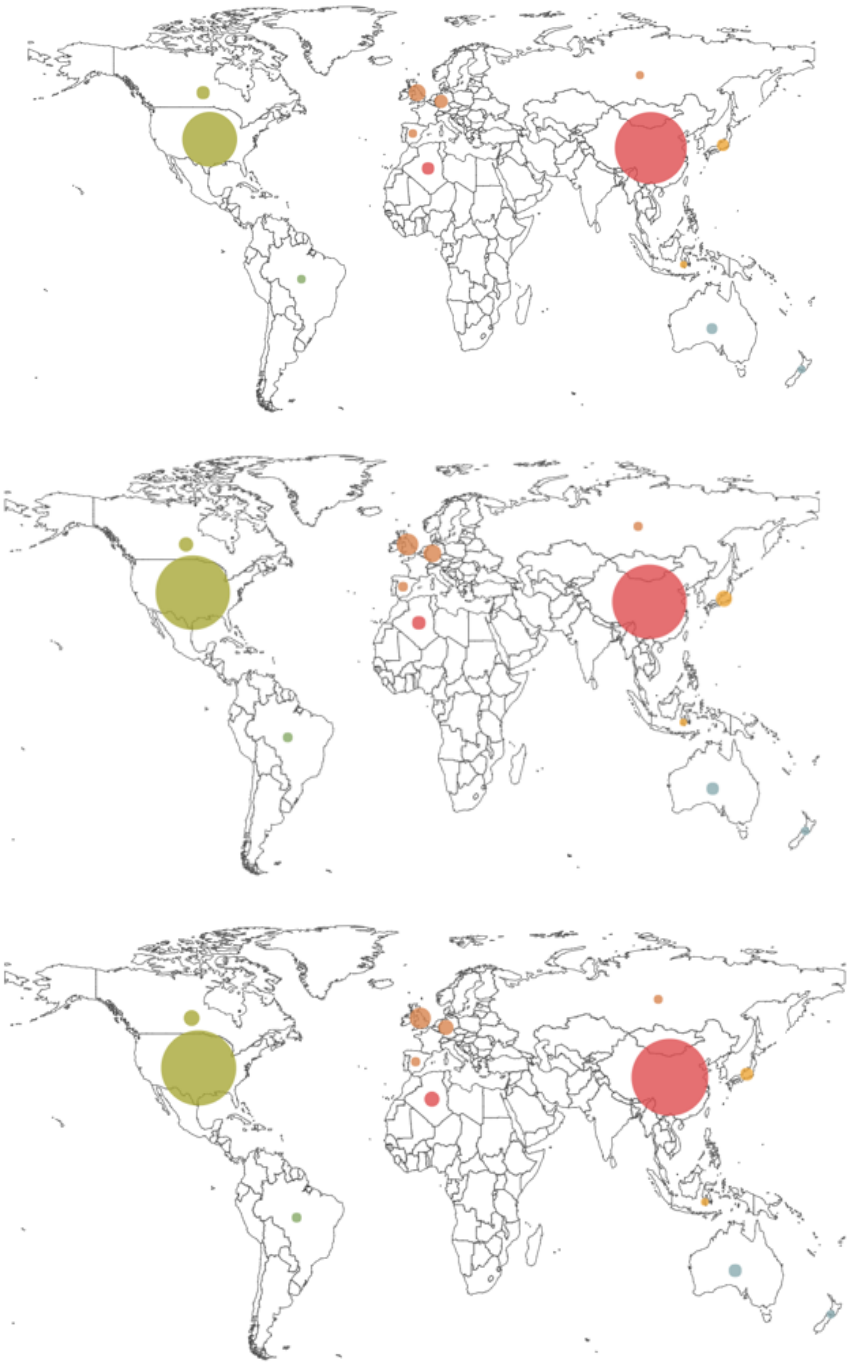

Figure 13: Thermodynamic Maps about English Global Distribution over Time

## 3.6 Gray Relational Prediction Analysis

The site selection of international offices is always related to the country's globalization level,

development elements and language impact. To quantify the three factors and make the prediction more succinct, we decide to use three indicators to denote the factors.

| Factor | Globalization Level | Development Elements | Language Impact |
|---|---|---|---|
| Indicator | Globalization Index (GI) | Global Competitiveness Index (GCI) | Language Indicator (LanI) |

Table 10: Indicators for Site Selection

Among the three indicators, GI and GCI are universal indicators while 'LanI' is an indicator we define by ourselves. We assume language indicator to be the sum of products of two elements, score of the language and proportion of people speaking the language in a certain country. We define Language Index in this way because we assume the language status of the place where international offices are built should be varied or influential (The language spoken there has a high language vitality).

We score languages according to their respective ranks.

| **Language** | **Rank** | **Score** | **Language** | **Rank** | **Score** |
|---|---|---|---|---|---|
| Mandarin | 1 | 10 | Malay | 6 | 5 |
| English | 2 | 9 | Russian | 7 | 4 |
| Hindustani | 3 | 8 | Portuguese | 8 | 3 |
| Spanish | 4 | 7 | Japanese | 9 | 2 |
| Arabic | 5 | 6 | German | 10 | 1 |

Table 11: Rank and Score of Languages

By assigning 9 scores to English, we can make sure that most people of the selected countries can speak English. Language Index can be described as

$$LanI_{Cont}^{Yr} = \sum_{i=1}^{n} (Scor_{Lan\ i}^{Yr} \times Propt_{Cont,Lan\ i}^{Yr})$$

We define the Comprehensive Index as an indicator to decide which country to choose from.

$$ComI_{Cont}^{Yr} = (GI_{Cont}^{Yr}, GCI_{Cont}^{Yr}, LanI_{Cont,Lan}^{Yr})$$

Because of the complex relationship among the tree indicators, we choose Gray Relational Prediction Analysis to confirm the selection system. Firstly, the index weight should be obtained. The concrete steps are as follows.

**Step 1**: Determine the reference data series as an ideal comparison standard. In our system, the largest value of each metric constitutes the reference sequence.

$$ComI_{Cont\ 0}^{Yr} = (ComI_{Cont\ 0}^{Yr}(1), ComI_{Cont\ 0}^{Yr}(2), \dots, ComI_{Cont\ 0}^{Yr}(m))$$

$$= (max\ (GI_{Cont}^{Yr}), max\ (GCI_{Cont}^{Yr}), max\ (LanI_{Cont,Lan}^{Yr}))$$

**Step 2:** Calculate the absolute difference value of each parameter sequence and the reference sequence of the corresponding elements as $\Delta_i(j) = |\, ComI_{Cont\ i}^{Yr}(j) - ComI_{Cont\ 0}^{Yr}(j)|$. In the formula, $i$ represents the number of candidate sites; $j$ is the number of metrics; $ComI_{Cont\ 0}^{Yr}(j)$ means the reference data in the metric.

**Step 3**: Calculate the correlation coefficient as

$$f_i(j) = \frac{min(\,\Delta_i(k)\,) + \rho \times max(\,\Delta_i(k)\,)}{\Delta_i(k) + \rho \times max(\,\Delta_i(k)\,)} \quad (\,\rho = 0.5\,)$$

**Step 4**: Calculate the correlation degree.

$$r_j = \frac{1}{n}\sum_{i=1}^{n} f_i(j)$$

**Step 5**: Calculate the weight of each measure standard.

$$r_j' = \frac{r_j}{r_1 + r_2 + \cdots + r_m}$$

**Step 6**: Build the evaluation model.

$$Z_i = r_1'\, x_i(1) + r_2'\, x_i(2) + \ldots + r_m'\, x_i(m)$$

To take all of the top ten languages into account and to make the offices cover more continents, we choose the UK, Singapore, Russia, Indonesia, India, Spain, Japan, Brazil and Germany as candidates. Based on the data we got above, we can solve the model and obtain the following results.

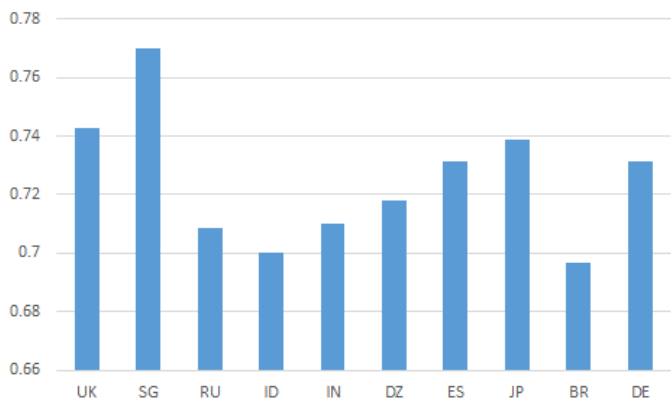


Figure 14: Results of Gray Relational Prediction Analysis

As shown in the figure above, **countries whose Complex Indicators rank in the top six are selected to build new international offices, including Singapore, the UK, Algeria, Japan, Germany and Spain.** The selected countries are painted orange on the map while the countries which already have an office are painted green.

**In the short term**, the site selection of international offices is a better result by taking the above factors into consideration. **In the long term**, due to the change of the number of language use and the influence of other factors such as economy, politics, culture, these results will change.

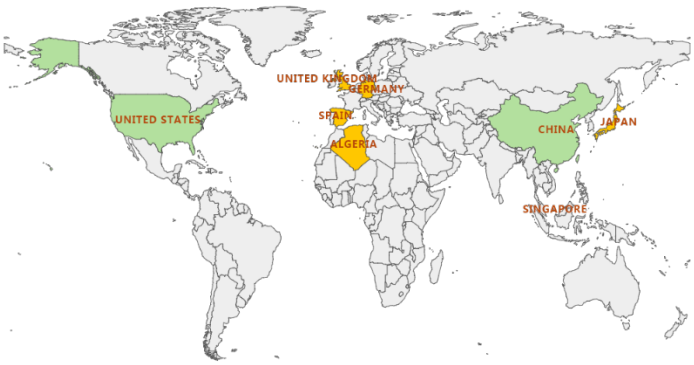


Figure 15: Site Selection

## 3.7 Maximum Minimum Distance Algorithm

Considering the changing nature of global communications, and in an effort to save client company resources, we can open less than six international offices. For countries sharing similar language features, we assume they can easily communicate on the Internet or by phone, so there is no need to open offices in each of these countries. We use a vector to describe the language feature.

$$\overrightarrow{f_{Cont}} = (\omega_{lan\ 1}, \omega_{lan\ 2}, \omega_{lan\ 3}, \omega_{lan\ 4}, \omega_{lan\ 5}, \omega_{lan\ 6}, \omega_{lan\ 7}, \omega_{lan\ 8}, \omega_{lan\ 9}, \omega_{lan\ 10})$$

$\overrightarrow{f_{Cont}}$ represents the language feature of country 'Cont'. $\omega_{lan\ i}$ represents the proportion of people in the country 'Cont' speaking language 'lan_i'. We select eight countries including Singapore, the UK, Spain, Algeria, Japan, Germany China, and the USA which already have international offices.

The Maximum Minimum Distance Algorithm is a heuristic clustering algorithm, which is based on the calculation of Euclidean distance. The algorithm chooses the farthest object as the new clustering center.

**1. Steps**

**Step 1**: Arbitrary choose a sample from the N samples as the first cluster center $Z_1$.

**Step 2**: Select the farthest sample from the first cluster center $Z_1$ as the second cluster center $Z_2$.

**Step 3**: Calculate the distance between a sample $x_i$ and $Z_1$ as well as the distance between the sample and $Z_2$. Compare the two distances and denote the smaller one by

$$d_{ij} = || x_i - Z_j ||, \qquad j = 1, 2$$

$$d_i = \min [\, d_{i1}, d_{i2} \,], \qquad i = 1, 2, \dots, N$$

**Step 4**: If $d_i = \max \{ \min[\, d_{i1}, d_{i2} \,] \} > \theta \times || Z_1 - Z_2 ||$ ( $\theta = 0.5$ ), then the corresponding sample $x_i$ is used as the third cluster center $Z_2$. And then the algorithm goes to the next step to determine whether there is a new cluster center. If there is not, then the algorithm goes to the seventh step.

**Step 5**: Assume that there are five clustering centers. Calculate the distances from each sample to each cluster center. Compare the distances and denote the largest one by

$$d_i = \max \{ \min [\, d_{i1}, d_{i2}, \dots, d_{ik} \,] \}$$

**Step 6:** If $d_i > \theta \times || Z_1 - Z_2 ||$ ( $\theta = 0.5$ ), then the corresponding sample $x_i$ is used as cluster center $Z_{k+1}$. Then the algorithm recycles to determine whether there is a new cluster center. If there is not, then the algorithm goes to the seventh step.

**Step 7:** When there is no longer a new cluster center, divide the sample set into clusters based on minimum distance principle.

**2. Results**

| Cluster | Country |
|---|---|
| 1 | Algeria |
| 2 | Japan |
| 3 | Germany |
| 4 | Spain |
| 5 | the USA<br>Singapore (omitted)<br>the UK (omitted) |
| 6 | China |

Table 12: Clustering Result

We can conclude from the clustering result that Singapore and the UK both share a similar language feature with the USA. Since there has already been an international office in New York City, the office can easily communicate with people in Singapore and the UK. So the construction of offices in the two countries can be omitted. **We can just open four more offices** which will help the company not only expand its scale but also save company resources.

# 4 Model Assessment

## 4.1 Use Residual Graph to Analyze Error

The residual graph can be used to judge whether the regression model can match the original data. The horizontal ordinate represents the identifier of the year, such as Case Number 1 represents the Year 2013.

From the residual graph, we can see the distance between residuals of data and the zero point. When the confidence intervals of residuals contain zero points, this indicates that the regression models can match the original data well. As the following figure shows, the regression model we establish can match the original data pretty well.

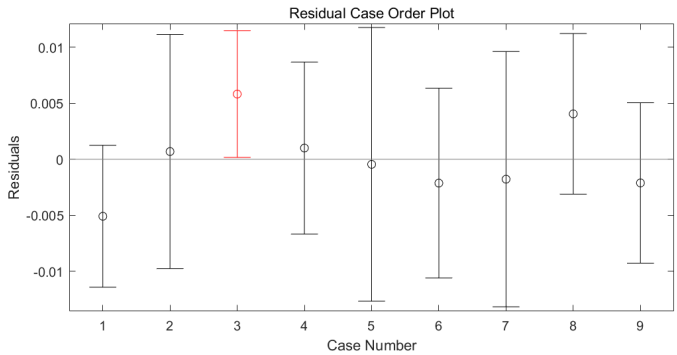


Figure 16: Residual Graph

## 4.2 Gray Relational Prediction Analysis

**1. Comparison of Actual Value and Prediction Value of Language Vitality of English**

| Year | Predicted value | True value | Error |
|---|---|---|---|
| 2008 | 1.2202 | 1.2122 | 0.0100 |
| 2011 | 1.1796 | 1.1885 | 0.0089 |
| 2013 | 1.1647 | 1.1520 | 0.0127 |

Table 13: Error Comparison

**2. Reference Table of Precision Test**

| Accuracy grade | Relative error |
|---|---|
| Excellent | 0.01 |
| Good | 0.05 |
| Qualified | 0.1 |
| Unqualified | 0.2 |

Table 14: Reference Table of Precision Test

It can be seen from the above table that the deviation between the prediction value and real value is small enough. The precision level of the model is close to the excellent level which means the model can be used for prediction.

# 5 Strengths and Weaknesses

## 5.1 Strengths

- The model takes the country as the research unit and takes the differences in the situation of different countries into account.
- Consider the important influencing factors as far as possible, and effectively predict the time and space changes of various language users.

## 5.2 Weaknesses

- There is not sufficient data to support the regression analysis to predict the universality of the model.
- The indicators considered are not perfect, and there are many factors that influence language change. The judgment matrix in AHP is more subjective.
- When calculating the language using the model of three indicators with less amount of data by the grey prediction, it is actually a kind of time series with some uncertain rules, so the prediction may have a deviation with the real situation.
- In the second part, ten candidate countries were selected to simplify the calculation, and the selection process was more subjective.

# 6 Memo

For the selection of locations around the world that can serve as international offices, we have considered countries on all continents as an alternative to expanding as much as possible. Based on the analysis of various languages, we finally determined ten candidate sites for the ten major languages.

Meanwhile, the Ranking of Globalization Index Report released recently by the KOF Swiss Economic Institute can tell us the ability and status of internationalization for distinct countries in the world, which is an important determinant of the selection for international offices. In addition, the Global Competitiveness Report released recently by the World Economic Forum has also delivered the key information to us. The report is released every year in order to evaluate the competitiveness of each country by twelve pillars including the infrastructure, education, market efficiency, macroeconomic framework, innovation and so on, which directly decide how competitive to be engaged in business.

According to the level of globalization, language diversity, and enterprise operation environment, through the analysis of large amounts of data, we finally chose Singapore, the UK, Algeria, Japan, Germany, and Spain for the construction of international offices. The rank and score derived from our model are as follows.

| Rank | Country | Score |
|---|---|---|
| 1 | Singapore | 0.7698 |
| 2 | the UK | 0.7429 |
| 3 | Japan | 0.7387 |
| 4 | Spain | 0.7315 |
| 5 | Germany | 0.7312 |
| 6 | Algebra | 0.7182 |

Table 15: Rank and Score of Locations

We believe that we can realize low cost but high return, because the six countries have high popularity of English or other various languages with positive development and performance. We are able to make reasonable use of the resources of these countries to develop and expand our business, and establish frequent exchanges with the international community to make new progress.

# 7 References

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